

REZA TALANEH

Mechanical Engineer

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SUMMARY

Mechanical engineer with a MSc in Applied Mechanical Engineering and research experience in computational mechanics, finite element analysis, and smart materials. My research and industry experience have given me a strong foundation in analytical modeling, numerical simulation, and product development. I am seeking a PhD position where I can contribute to research in computational mechanics, mechanical design, advanced manufacturing, or sustainable product development while addressing real engineering challenges.

RESEARCH INTERESTS

- Applied mechanical design and structural analysis for industrial systems
- Finite element modeling of nonlinear material and structural behavior
- Constitutive and computational modeling of smart materials and shape memory alloys
- Design for manufacturability and industry-oriented product development
- Computational and applied continuum mechanics

EDUCATION

Babol Noshirvani University of Technology (THE Eng. Rank 2026: 501–600) **Babol, Mazandaran, Iran**

MSc in Applied Mechanical Engineering

Sep 2013 - Feb 2016

- Graduated with a **GPA of 18.45/20**, ranking **2nd** in a cohort of 24 students.
- Thesis Title: "**Analytical and finite element study of super elasticity behavior and shape memory effect of a Shape memory alloy thick-walled cylinder**" Supervisor: Prof. R. A. Alashti; Advisor: Prof. A. Fathi
- **Notable Courses:** Plasticity (20/20) • Continuum Mechanics (18.25/20) • Finite Element Method I (19.2/20) • Advanced Numerical Computations (18.1/20) • Optimal Design (19.7/20)

BSc in Mechanical Engineering

Sep 2009 - Sep 2013

- Graduated with a **GPA of 17.33/20**, ranking 2nd in a cohort of 63 students.
- Thesis Title: "**Finite Element Analysis of Creep Behavior in Orthotropic Sandwich Beams with a Viscoelastic Core under Bending**" Supervisor: Prof. R. A. Alashti
- **Notable Courses:** Statics (20/20) • Materials Science (19.6/20) • Mechanism Design (20/20) • Dynamics of Machinery (19.25/20) • Engineering Design Methods (20/20) • Computer Aided Machine Design (18/20)

RESEARCH EXPERIENCE

Babol Noshirvani University of Technology

Babol, Mazandaran, Iran

Graduate Researcher - Nonlinear FE & Constitutive Modelling of Shape Memory Alloys

Sep 2013 - Feb 2016

- Reduced the 3D Boyd–Lagoudas SMA constitutive model to plane-stress boundary value problems for thick-walled Ni-Ti cylinders under internal pressure and combined pressure–torsion loading, and solved the resulting nonlinear coupled equations using a custom MATLAB finite-difference solver.
- Validated the model against two published solutions and 3D Abaqus FE simulations using an existing Fortran UMAT, and characterized martensitic phase evolution and stress redistribution under varying pressure, torsion, and temperature.

Undergraduate Researcher - Long-Term Creep Behaviour of Sandwich Structures

May 2013 - Sep 2013

- Modeled an orthotropic composite sandwich beam with a viscoelastic core in ANSYS (SOLID186, implicit creep formulation), evaluating time-dependent creep strain and tip displacement under bending.

PUBLICATIONS

- Alashti, R.A., Talaneh, R., Ezami, A.H. **“Analytical and Finite Element Study of Superelasticity Behavior and Shape Memory Effect of SMA Thin-Walled Cylinders under Torsional Loading.”** *Proceedings of the 23rd International Conference on Mechanical Engineering (ISME2015), Tehran, Iran, 2015. (Oral presentation)*
- Alashti, R.A., Ezami, A.H., Talaneh, R. **“Analytical and Numerical Modeling of Pseudoelastic and Shape Memory Behavior of Shape Memory Alloys under Uniaxial Tension.”** *Proceedings of the 1st National Conference on Applied Mechanics, Azad University, Tabriz Branch, Tabriz, Iran, 2014.*

TEACHING EXPERIENCE

Babol Noshirvani University of Technology

Babol, Mazandaran, Iran

Teaching Assistant – Mechanics of Materials Laboratory

Feb 2014 - Feb 2015

- Assisted in delivering more than 20 undergraduate Mechanics of Materials laboratory sessions, supporting experimental instruction and student learning.

PROFESSIONAL EXPERIENCE

ABCHECK CO (Manufacturer of Agricultural & Industrial Plastic Blown Films) Engineering Manager

Mahmudabad, Mazandaran, Iran
Oct 2017 - Present

- Leads multidisciplinary manufacturing teams, applying engineering KPIs and data-informed decision-making across product lines.

Design and Development Manager

Sep 2016 - Sep 2017

- Led engineering design, prototype development, testing, and validation.

Research and Development Supervisor

Sep 2013 - Sep 2016

- Supervised new-product development and manufacturing-process improvement.

INDEPENDENT DESIGN PROJECTS

Self-directed mechanical design projects undertaken outside of academic or employer scope:

- **Blown Film Dual-Lip Air Ring:** Reverse-engineered and modeled an industrial dual-lip air ring used in blown film extrusion, emphasizing manufacturability and mechanical functionality.
- **Two-Stage Parallel-Shaft Gearbox Design:** Complete gearbox design including transmission calculations, shaft design, bearings selection, housing development, motion verification, and manufacturing drawings.
- **Blueprint-to-CAD Reconstruction & Structural FEA:** Reconstructed production-ready CAD models from engineering drawings and performed structural FEA to evaluate stress distribution and mechanical behavior.

CORE COMPETENCIES

Engineering: Product Development • Mechanical Design • CAD • DFM • FEA • Engineering KPIs • Data-Informed Decision-Making • Cross-Functional Collaboration

Software: SolidWorks • ANSYS • Abaqus • MATLAB • Power BI • Azure DevOps

LANGUAGES

English: PTE Academic Overall 90/90 (CEFR C2) • German: Beginner • Persian: Native

PROFESSIONAL MEMBERSHIP

International Association of Engineers (IAENG)

REFERENCES

Available upon request